

Comprehensive—Volume + Area + Fraction

Cross-Topic integration (T1 · T5 · T9) · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Units 7-8 + Modern Education 5B Unit 18-20

Core traps: ● High Frequency

Sub-test cross-topic comprehensive questions, accounting for about 18-22% of Paper 1 **Prerequisite knowledge:**

Class 21 (Circle + Section + Origami) · Class 22-23 (Fundamentals of Volume + Application) · Class 24-26 (Multiplication and Division of Fractions) · Class 27 (Division of Decimals)

Objectives of this class: ❶ Comprehensive application of volume and area formulas ❷ Application of fractions in volume ❸ Identification and avoidance of cross-topic traps

Our goals: ❶ Comprehensive application of volume and area formulas ❷ Application of fractions in volume ❸ Identification and avoidance of cross-topic traps

Student Name:

Class:

Date:

Time Spent:

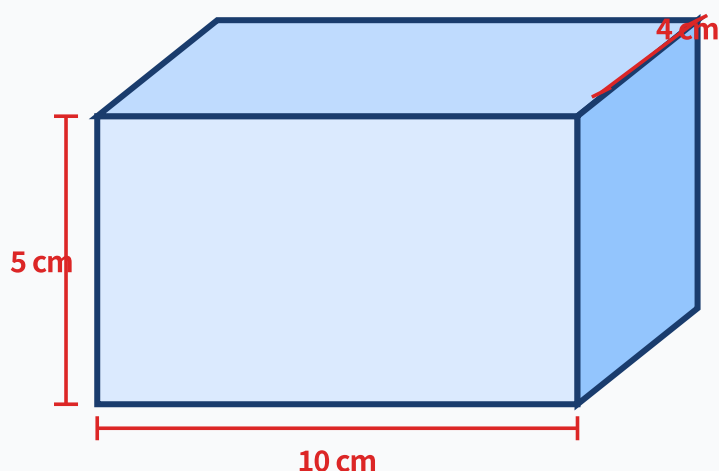
I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	The cuboid is 6 cm long, 4 cm wide and 5 cm high. Volume = ?	Basic	
2	Triangular base 8 cm, height 6 cm. Area = ?	Basic	
3	The parallelogram has a base of 12 cm and a height of 5 cm. Area = ?	Basic	
4	The base area of a cuboid is 24 cm ² and the height is 3 cm. Volume = ?	Advanced	
5	The volume of a cuboid is 120 cm ³ , its length is 6 cm, and its width is 5 cm. Seek high.	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Comprehensive application of volume + area formula ● SSPA

- ① **Volume formula (cuboid/cube):** $V = \text{length} \times \text{width} \times \text{height}$; $V = \text{base area} \times \text{height}$
- ② **Area formula review:** Triangle $A = (\text{base} \times \text{height}) \div 2$ • Parallelogram $A = \text{base} \times \text{height}$ • Trapezoid $A = (\text{upper base} + \text{lower base}) \times \text{height} \div 2$
- ③ **Key distinction:** The area is the size of the "surface" (cm²/m²), and the volume is the size of the "space" (cm³/m³)
- ④ **Composite graphics strategy:** Divide into basic graphics → Find area/volume separately → Add or subtract
- ⑤ **Unit trap:** The base and height units must be consistent; the volume unit is cubic (³), not square (²)!



$$V = \text{length} \times \text{width} \times \text{height} = 10 \times 4 \times 5 = 200 \text{ cm}^3$$

❑ Example of trap detonation (the most important demonstration in this class)

A rectangular container has a trapezoidal base (upper base 4 cm, lower base 6 cm, height 5 cm), and the height of the container is 10 cm. Find the volume of the container.

✗ Common mistakes (70% students)

✓ Correct solution

$$V = (4+6) \times 5 \times 10 = 500 \text{ cm}^3$$

Forget about the area of a trapezoid $\div 2$! First find the base area = $(4+6) \times 5 \div 2 = 25 \text{ cm}^2$, then $\times 10 = 250 \text{ cm}^3$

$$V = 250 \text{ cm}^3$$

① Base area = $(4+6) \times 5 \div 2 = 25 \text{ cm}^2$ ② $V = 25 \times 10 = 250 \text{ cm}^3$

Tip: "First determine the surface and determine the body, and distinguish the basis of the formula; divide the area of a trapezoid by two, and then multiply the volume by the height; the unit is cubed, and the area is raised to the second power."

⚠ The most frequent error: Forget the area of the trapezoid $\div 2$, use (upper base + lower base) $\times$ height directly as the area. Remember that the trapezoidal formula must be divided by 2!

⚠ The second most common error: the unit of volume is written as cm^2 instead of cm^3 - area is square and volume is cubic!

Knowledge Point - Worked Examples (write out complete steps: ①findbasearea ②findvolume ③mark unit)

#	Question	Difficulty	Working Space
Example 1	The side length of a cube is 5 cm. Find: (a) the area of a face (b) the volume		
Example 2	The cuboid is 8 cm long, 6 cm wide, and 4 cm high. Find the area and volume of the base.		
6	A rectangular parallelepiped with a square base (side length 7 cm) and height 10 cm. Find the volume.		
7	A rectangular container has a parallelogram base (base 8 cm, height 5 cm) and the height of the container is 12 cm. Find the volume.		
8	A rectangular parallelepiped has a base area of 36 cm^2 and a height of 9 cm. If the height is increased by 3 cm, how much will the volume increase?		
9	The picture below consists of two cuboids (stacked one above the other). Upper layer: $3 \times 3 \times 4 \text{ cm}$; lower layer: $5 \times 5 \times 6 \text{ cm}$. Find the total volume. Tip: Calculate separately and then sum.		
10	The side length of one cube is 2 times the length of the other cube. How many times the volume of the large cube is the volume of the small cube?		

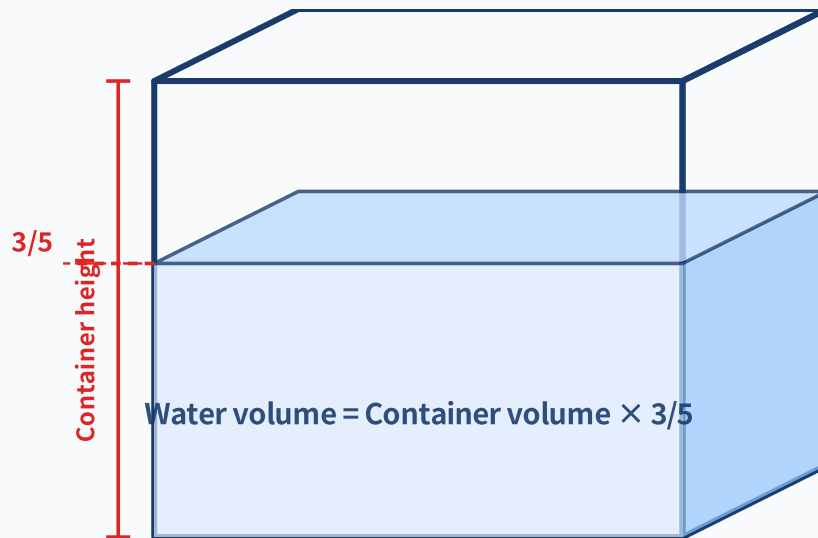
Knowledge Point 2: Application of Fractions in Volume SSPA Required Exam

Four ways to consider fractions in volume:

① **Side length contains fractions:** Length = $2\frac{1}{2} \text{ cm}$ $\rightarrow$ Convert to false fraction and multiply: $V = \frac{5}{2}$ Container volume

② **Fraction of volume:** The container is packed $\frac{3}{4}$ of water $\rightarrow$ volume of water = volume of container $\times \frac{3}{4}$

- ③ **Find the volume of the parts and find the whole:** known $\frac{2}{5}$ The volume is $40 \text{ cm}^3 \rightarrow \text{whole} = 40 \div \frac{2}{5} = 100 \text{ cm}^3$
- ④ **Scaling:** Length enlargement $1\frac{1}{2}$ times $\rightarrow$ volume enlargement $(\frac{3}{2})^3 = \frac{27}{8}$ times



①

Side length includes fraction

Mixed fractions $\rightarrow$ improper fractions and then multiply to find the volume

②

Find the fraction

Volume $\times$ Fraction = Partial Volume

③

Rounding the known parts

Part $\div$ Fraction = Whole Volume

④

scaling

Side length $\times$ k Volume $\times$ k^3

example

Example 3: A cuboid is $2\frac{1}{2}$ cm long, 4 cm wide, and 3 cm high. Find the volume.

example

Example 4: A water tank has a volume of 80 L and is filled with $\frac{3}{5}$ water. What is the volume of water in L?

✗ Common mistakes

$$2\frac{1}{2} \times 4 \times 3 = 2.5 \times 12 = 30$$

After converting mixed numbers to improper fractions, the numerator and denominator are processed separately: $52 \times 4 \times 3 = 30 \checkmark$ (This question happens to be correct), but it is easy to get confused in complicated situations!

✓ Correct approach

$$V = \frac{5}{2} \times 4 \times 3 = 30 \text{ cm}^3$$

$$\textcircled{1} 2\frac{1}{2} = \frac{5}{2} \quad \textcircled{2} \frac{5}{2} \times 4 \times 3 = \frac{60}{2} = 30$$

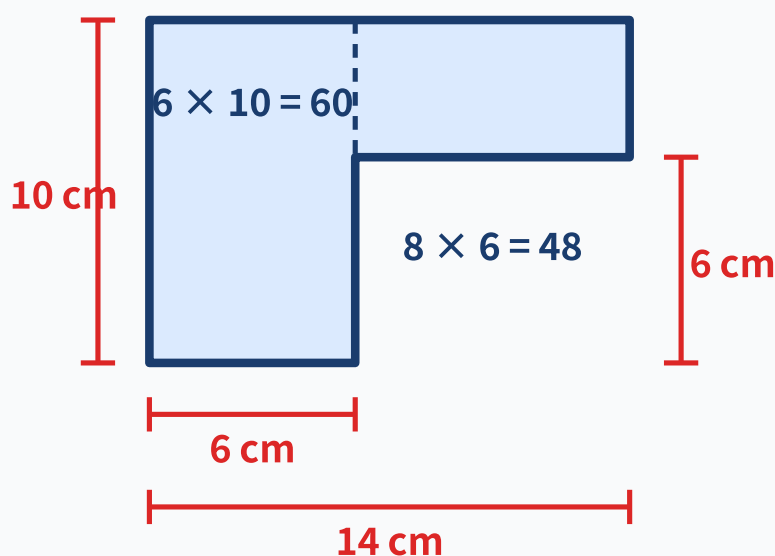
#	Question	Difficulty	Working Space
11	The side length of the cube is $1\frac{1}{2}$ cm. Find the volume. (Answer in fractions)		
12	A water tank has a volume of 60 L and contains $\frac{2}{3}$ water. How many liters of water is there?		
13	A fish tank $\frac{3}{4}$ has 45 L of water. What is the volume of the entire fish tank in L?		
14	The length of the cuboid is $\frac{3}{2}$ m, the width $\frac{2}{3}$ m, and the height is 2 m. Find the volume. (Answer with m^3)		
15	After the side length of the cube is enlarged by $1\frac{1}{3}$ times, how many times is the new volume the original? (Answer in fractions)		

Knowledge Point 3: Comprehensive Application—Volume + Area + Fraction Three-in-One

SSPA Advanced

Strategies for solving cross-topic comprehensive questions (three-step method for reading questions):

- ① **Circle all the numbers and units in the question**- Clarify which are areas, which are volumes, and which are fractions
- ② **Judge the order of operations**- First find the area (base area) → then find the volume → finally deal with the fraction ratio
- ③ **Check the consistency of the units**—m and cm cannot be multiplied directly! Convert to the same unit first



$$\text{Total area} = 60 + 48 = 108 \text{ cm}^2$$

example

Example 5: A rectangular water tank with a rectangular base (1 m long, 50 cm wide) and 80 cm high. The water tank contains $\frac{3}{5}$ water. Find the volume of water (answer in L, 1 L = 1000 cm^3).

example

Example 6: The side length of cube A is 6 cm, and the side length of cube B is $\frac{2}{3}$ of A. What fraction of the volume of B is A?

 **Tip: "Unify the units first, then the area and then the volume; multiply the fractions last, and avoid all traps."**

 **High frequency trap: m and cm are mixed up - 1 m = 100 cm, but 1 m² = 10000 cm², 1 m³ = 1,000,000 cm³!**

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
16	The base area of a cuboid is 25 cm ² and the height is 8 cm. What is the volume $\frac{3}{4}$ in cm ³ ?		
17	A fish tank is 50 cm long, 40 cm wide, and 30 cm high. (a) Find the volume of the fish tank (b) If water is filled to the height of $\frac{4}{5}$ Height, what is the volume of water in L?		
18	The side length of a cube water tank is 20 cm. Existing water 4000 cm ³ . What fraction of the tank's volume does water occupy? (Answer with the simplest fraction)		
19	The cuboid is 1.5 m long, 0.8 m wide, and 0.6 m high. (a) Volume = ? m ³ (b) If the space of $\frac{5}{8}$ has been used, how much m ³ is the remaining space?		
20	A cylinder has a base area of 78.5 cm ² and a height of 10 cm. If the height is reduced by $\frac{1}{5}$, what is the new volume? (Hint: new high = $10 \times (1 - \frac{1}{5})$, what is the new volume? (Hint: new high = $10 \times (1 - \frac{1}{5})$)		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
21	The cuboid is 10 cm long, 5 cm wide, and 4 cm high. Find the volume.		
22	The side length of the cube is 8 cm. Find the volume.		
23	Triangular base 12 cm, height 9 cm. Find the area.		

24	The upper base of the trapezoid is 4 cm, the lower base is 8 cm, and the height is 5 cm. Find the area.		
25	The volume of a cuboid is 72 cm^3 and the base area is 12 cm^2 . Seek high.		

Advanced layer (total 5 questions, choose do)

#	Question	Difficulty	Working Space
26	A cuboid has a square base (sides 6 cm) and a height 15 cm. Find the volume.		
27	Water tank volume 120 L, used SEP . How much L space is left? $\frac{5}{6}$. How much L space is left?		
28	The side length of the cube is $1\frac{1}{3}$ cm. Find the volume. (Answer with improper fractions)		
29	Cuboid A: length 6 cm, width 4 cm, height 5 cm. Cuboid B: All sides are twice as long as A. How many times the volume of B is A?		
30	A combined solid: the lower cuboid is $10 \times 8 \times 5$ cm, and the upper cube with side length 4 cm is placed in the center. Find the total volume.		

challenge layer (total 3 questions, choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
31	A rectangular container has an interior length of 30 cm, a width of 20 cm, and a height of 15 cm. After placing a stone, the water level rises by 2 cm. What is the volume of the stone in cm^3 ? Tips: Drainage method - stone volume = bottom area $\times$ water level rise height		
32	The side length of the cube is $\frac{3}{4}$ m. Find (a) the area of a surface (answer in m^2) (b) the volume (answer in m^3) (c) how many times the volume is the area of the base?		
33	A trapezoidal cylinder: the base is trapezoidal (upper base 5 cm, lower base 9 cm, height 4 cm), and the height of the cylinder is 10 cm. Find the volume. Tips: $V = \text{base area} \times \text{height} = \text{trapezoid area} \times \text{cylinder height}$		

IV.applicationquestionspecial topic training

#	Question	Difficulty	Working Space
34	A rectangular pool is 8 m long, 5 m wide, and 2 m deep. (a) What is the volume of the pool in m^3 ? (b) If the water depth is only SEP , what is the volume of water in m^3 ? $\frac{3}{4}$, what is the volume of water m^3 ?		

35	A rectangular container is 12 m long, 2.5 m wide, and 2.8 m high. (a) What is the volume of the container in m ³ ? (b) The loaded goods occupy $\frac{5}{7}$ of the container, how many m ³ of goods can be loaded in the remaining space?		
36	A cube water tank has a side length of 40 cm. 8000 cm ³ of water is injected every minute. (a) What is the total volume of the tank in cm ³ ? (b) How many minutes does it take to fill the water tank? (c) After 5 minutes, what fraction of the water will be in the tank?		
37	A rectangular fish tank is 60 cm long, 30 cm wide, and 40 cm high. After first pouring 36 L of water and adding some stones, the water level rose by 5 cm. (a) 36 L What is the height of the water in the tank in cm? (1 L = 1000 cm ³) (b) What is the total volume of the stone in cm ³ ? (c) What is the final water depth in cm?		

V. Ultimate challenge area

#	Question	Difficulty	Working Space
 1	The length of a cuboid is 2 times the width, and the breadth is $1\frac{1}{2}$ times the height. If the height is 4 cm, find: (a) What is the width? (b) What is the length? (c) What is the volume? (d) What is the surface area? (Surface area = $2 \times (\text{length} \times \text{width} + \text{length} \times \text{height} + \text{width} \times \text{height})$)		
 2	A cube has a side length of 10 cm. A small cube with a side length of 2 cm is dug out in the center of each of its six faces (digging through to the opposite side). Find the volume of the remaining solid. Tips: Note that the perforations in the three directions will overlap, be careful not to deduct repeatedly!		
 3	A trapezoidal cylindrical pool: the bottom is trapezoidal (upper bottom 3 m, lower bottom 7 m, height 4 m), and the pool depth is 2 m. 32 m ³ of water available. (a) What is the total volume of the pool in m ³ ? (b) What fraction of the total pool is the existing water? (Answer in simplest fractions) (c) How many m ³ of water must be injected to fill it completely?		

Cross-topic integration: volume + area + fraction, a universal problem-solving framework

essential for score examinations

When encountering comprehensive questions, deal with them in the following order:

- ① **Read the questions:** Circle all the numbers, mark the units (cm/m/L/cm³/m³), and find out "a fraction"
- ② **Unify the units:** Unify all lengths into the same unit (all cm or all m)
- ③ **Area first, then volume:** If you need the base area → use the corresponding formula (rectangle/triangle/trapezoid/parallelogram)

- ④ **Processing fractions:** Volume $\times$ fraction = volume of part; part $\div$ fraction = whole
- ⑤ **Check the answer:** Are the units correct? Are fractions reduced? Is the answer reasonable?

VI. Class after homework

Basic must-do questions (total 5 questions, must Show full working)

#	Question	Difficulty	Working Space
H1	The cuboid is 7 cm long, 5 cm wide, and 3 cm high. Find the volume.		
H2	The side length of the cube is 9 cm. Find the volume.		
H3	A cuboid has a base area of 28 cm^2 and a height of 6 cm. Find the volume.		
H4	Water tank volume 90 L. Installed $\frac{2}{5}$ water. How many liters of water is there?		
H5	A cuboid is 1.5 m long, 0.8 m wide, and 0.5 m high. Find the volume (answer in m^3).		

Advanced choose do question (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	The cuboid is $2\frac{1}{2}$ cm long, 4 cm wide and 3 cm high. Find the volume.		
H7	A cube water tank has a side length of 25 cm. Pour 8 L of water first, then add the stones, and the water level rises by 3 cm. What is the volume of the stone in cm^3 ? (1 L = 1000 cm^3)		
H8	Cuboid A: $6 \times 4 \times 5 \text{ cm}$. Cuboid B: All side lengths are $1\frac{1}{2}$ times that of A. How many times the volume of B is A? (Answer in fractions)		

VII. The Lesson core common errors summary

☒ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

#	common error	Correct Approach
1	Forget the area of the trapezoid $\div 2$ (upper base + lower base) $\times$ height, not the area of the trapezoid!	Trapezoidal area = (upper base + lower base) $\times$ height $\div 2$, you must also divide 2 when using it to calculate the base area.
2	The volume unit is written incorrectly SEP : cm^3 is written as cm^2 : cm^3 is written as cm^2	Area $\rightarrow$ square (2), volume $\rightarrow$ cubic (3). Check the index before each answer
3	m/cm conversion error SEP : $1\text{m}=100\text{cm}$ but $1\text{m}^3 \neq 100\text{cm}^3$: $1\text{m}=100\text{cm}$ but $1\text{m}^3 \neq 100\text{cm}^3$	$1\text{ m}^3 = 100 \times 100 \times 100 = 1,000,000\text{ cm}^3$. Three-dimensional conversion requires conversion in each dimension
4	The fractions are not reduced after multiplication : $\frac{6}{12}$ Take it directly as the answer	The last step is to check whether the numerator and denominator have common factors, which must be reduced to the simplest
5	"Part $\div$ Fraction = Whole" is reversed SEP : Use multiplication instead of division : Use multiplication instead of division	Given that $\frac{3}{5}$ is 60 $\rightarrow$ overall = $60 \div \frac{3}{5} = 100$ (not $60 \times \frac{3}{5}$)
6	The side length is enlarged k times $\rightarrow$ the volume is enlarged k times (Wrong!)	The side length is enlarged k times $\rightarrow$ the volume is enlarged k^3 times (cubic!)
7	Misunderstanding of drainage method SEP : Thinking that the rising height of the water level = the height of the stone : Thinking that the rising height of the water level = the height of the stone	The volume of the stone = the bottom area of the container $\times$ the rising height of the water level (not the height of the stone)

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 Related topics: L03 Area of parallelograms and triangles • L04 Area of trapezoidal polygon • L05 Area trap special project • Related topics: L25 Volume word problems • L26 Volume concept • L27 Composite three-dimensional drainage method